Singularities in soft matter: how rheology picks the pinch-off exponent

Pull a thread of shampoo apart between two fingers and watch what the last millisecond does. A column of water would thin smoothly into a cusp and separate, obeying the self-similar pinch-off law Eggers wrote down thirty years ago (Eggers, 1993). Shampoo does almost nothing of the sort. The thread stalls, elastic stress builds, a uniform cylindrical filament forms and thins exponentially in time, beads appear along it, and only then — sometimes — does it break. The Newtonian answer is a theorem (Eggers and Fontelos, 2015). The viscoelastic answer is an open zoo. Different constitutive models — Oldroyd-B, Giesekus, Johnson–Segalman, Rolie–Poly, soft-glassy rheology — either regularise the singularity, re-select its exponents, or forestall it entirely through an extensional necking instability that breaks the filament before a singular point can form (Fielding, 2011). This project charts, with direct numerical simulation carried all the way to breakup, which of those three fates each model selects, and where in (Wi, De, Ec) the boundaries lie.

TL;DR

Newtonian filament pinch-off has a textbook self-similar answer (Eggers, 1993; Eggers and Fontelos, 2015); viscoelastic pinch-off does not. Which of three fates a thread selects — a regularised (non-singular) Oldroyd-B exponential thinning (Entov and Hinch, 1997), a re-selected singular exponent at finite polymer extensibility (Giesekus, FENE-P), or collapse through an extensional necking instability that kills the filament before the singularity forms (Fielding, 2011; Hoyle and Fielding, 2016) — is set by the constitutive law, the finite-extensibility ingredient, and flow-curve non-monotonicity (Wilson and Fielding, 2006). Governing groups are the Ohnesorge number $Oh = \mu/\sqrt{\rho\gamma R_0}$, the Weissenberg number $Wi = \lambda\dot{\varepsilon}$, the Deborah number $De = \lambda/\tau_{\rm cap}$ with $\tau_{\rm cap} = \sqrt{\rho R_0^3/\gamma}$, and the elasto-capillary number $Ec = GR_0/\gamma$. You will benchmark the Newtonian Eggers solution and the Entov–Hinch exponential regime in Basilisk C, implement a ladder of constitutive models (Oldroyd-B (Entov and Hinch, 1997), Giesekus/FENE-P (Bhat et al., 2010), Johnson–Segalman (Wilson and Fielding, 2006), Rolie–Poly (Likhtman and Graham, 2003), soft-glassy rheology (Sollich et al., 1997)) in the log-conformation formulation (Fattal and Kupferman, 2004) to keep polymer stress stable at large Wi, sweep the (Wi, De, Ec) plane, and benchmark against the classical CaBER methodology (McKinley and Tripathi, 2000). Outputs: a regime map of viscoelastic singularity classes and an open-source CoMPhy-Lab workflow, co-supervised with Prof. Suzanne Fielding's group.

Description

Every fluid breaks, and how it breaks is set by its constitutive law at strain rates the neck selects as it thins: the mid-filament radius evolution $R_{\rm min}(t)$ fingerprints the rheology across a band of timescales that no single rheometer setpoint exposes. For a Newtonian thread the answer is textbook: a self-similar pinch-off with universal exponents, derived from the (η, γ, ρ) balance in a narrowing neck and verified to within a few percent in experiment (Eggers, 1993; Eggers and Fontelos, 2015). The viscoelastic case is an open zoo. Some polymer threads refuse to break at all over any experimentally accessible timescale. Others bead up into a chain of droplets connected by ever-thinner cylindrical ligaments (Bhat et al., 2010). Others collapse through an extensional necking instability well before the singularity forms, producing a heterogeneous filament that breaks at a chosen weakest point rather than at a generic self-similar cusp (Fielding, 2011; Hoyle and Fielding, 2016). Which of those

fates a fluid selects depends entirely on its constitutive law, and pinch-off offers one of the sharpest experimentally accessible tests of that law, precisely because the thinning neck sweeps through a continuous band of strain rates in a single run rather than interrogating the rheology at one imposed setpoint.

Four dimensionless groups govern the dynamics once the geometry and bulk viscosity are fixed. The Ohnesorge number $Oh = \mu/\sqrt{\rho\gamma R_0}$ sets viscous forces against the inertial–capillary scale and distinguishes inertial from viscous Newtonian pinch-off. The Weissenberg number $Wi = \lambda\dot{\varepsilon}$ compares polymer relaxation against the instantaneous deformation rate in the neck; it is not externally imposed but rises as the neck thins, and any constitutive model must be integrated forward until Wi passes through whichever finite-extensibility ceiling it imposes. The Deborah number $De = \lambda/\tau_{\mathrm{cap}}$ with capillary time $\tau_{\mathrm{cap}} = \sqrt{\rho R_0^3/\gamma}$ ranks relaxation against the thread's natural capillary-wave timescale. The elasto-capillary number $Ec = GR_0/\gamma$ weighs the polymer modulus G against Laplace pressure; in the exponentially thinning regime the thread radius follows $R(t)/R_0 \sim \exp(-t/3\lambda)$, and Ec sets the crossover at which that elastic regime begins (Entov and Hinch, 1997). A fifth, model-specific parameter — the retardation ratio $\beta = \eta_s/\eta$ or finite-extensibility ceiling $L_{\max}$ of the chosen constitutive law — is the one that actually selects the breakup scenario.

The constitutive ladder encodes the scientific question. Oldroyd-B isolates linear elasticity and predicts Entov–Hinch exponential thinning, which never reaches a singular point in finite time (Entov and Hinch, 1997): the singularity is *regularised*. Giesekus and FENE-P reinstate a finite polymer-extension ceiling; once the stretch saturates, the thread can pinch off again, but with a *re-selected* exponent that depends on the saturation law; this is the regime in which beads-on-a-string chains form (Bhat et al., 2010). Johnson–Segalman introduces a non-monotonic flow curve and admits shear-banded solutions; its relevance to filament breakup is not that shear bands form in extension (they do not), but that the same non-monotonicity that supports shear banding in simple shear drives an extensional necking instability whose signatures Prof. Suzanne Fielding's group has derived analytically (Fielding, 2011; Hoyle and Fielding, 2016; Wilson and Fielding, 2006). The Rolie–Poly model of entangled polymer melts adds tube dynamics, a convective-constraint-release mechanism, and a second relaxation timescale, so the thinning does not collapse onto a single exponential (Likhtman and Graham, 2003). Soft-glassy rheology introduces aging, yielding, and weak long-time memory; at the singularity it is unknown whether pinch-off resembles a yield-stress fluid, a power-law liquid, or something qualitatively different (Sollich et al., 1997). The project tests each of these predictions numerically, end-to-end from initial neck seeding down to the last resolvable radius, and places the boundaries between regularisation, re-selection, and necking-forestalled breakup as curves in the (Wi, De, Ec) plane.

The numerical challenge is that polymer-stress components grow exponentially during thinning, and conventional integration of the conformation tensor $\mathbf{C}$ loses positive-definiteness at moderate Wi (the high-Weissenberg-number problem). The log-conformation formulation evolves $\log \mathbf{C}$ instead, preserves positivity by construction, and remains stable through the entire exponential-thinning regime and into the re-selection regime (Fattal and Kupferman, 2004). Combined with Basilisk's adaptive mesh refinement targeted at the neck, it lets a single simulation resolve four decades of radial thinning without remeshing, and supports axisymmetric two-phase DNS up to and into the singular limit (Popinet, 2015; Sanjay, 2022). The classical Capillary Breakup Extensional Rheometer (CaBER) methodology (McKinley and Tripathi, 2000) gives a direct experimental benchmark: the mid-filament radius evolution predicted by the simulations maps one-to-one onto the CaBER observable, so every point on the regime map has a testable signature.

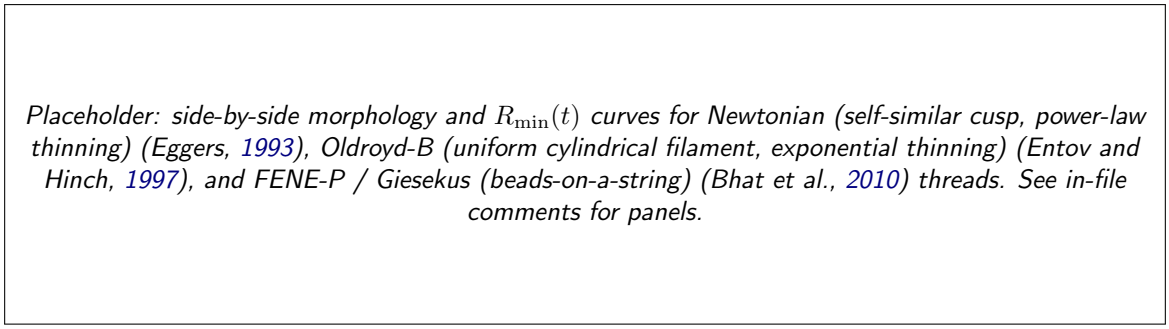


Figure 1: The phenomenological divide the project must explain. Newtonian pinch-off converges to a universal self-similar cusp with $R_{\min} \propto (t_c - t)$ exponents derived from the (η, γ, ρ) balance at the neck (Eggers, 1993; Eggers and Fontelos, 2015). Oldroyd-B thinning is exponential, $R_{\min}(t) \propto \exp(-t/3\lambda)$, and never reaches $R_{\min} = 0$ in finite time (Entov and Hinch, 1997): the singularity has been *regularised*. Models with finite polymer extensibility (Giesekus, FENE-P) saturate out of the exponential regime and break up again, but with a *re-selected* exponent characteristic of the saturation mechanism, producing beads-on-a-string morphology (Bhat et al., 2010).

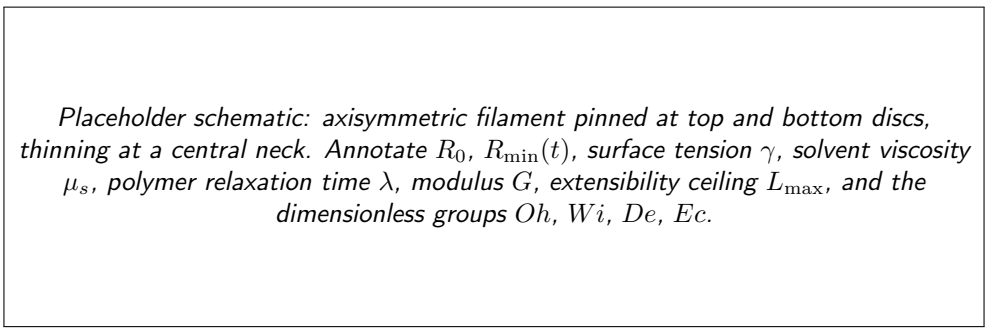


Figure 2: Idealised model problem. A liquid thread is pinned between two axisymmetric discs; depending on the protocol (imposed Hencky strain rate or post-step-stretch CaBER-like relaxation) the thread either elongates under an externally controlled $\dot{\varepsilon}$ or thins passively under capillary pressure (McKinley and Tripathi, 2000). The Basilisk simulations resolve the full axisymmetric two-phase flow with adaptive mesh refinement concentrated at the neck (Popinet, 2015; Sanjay, 2022), and the polymer conformation is advanced in the log-conformation formulation (Fattal and Kupferman, 2004) to keep $\mathbf{C}$ positive-definite through the exponential-thinning regime and beyond.

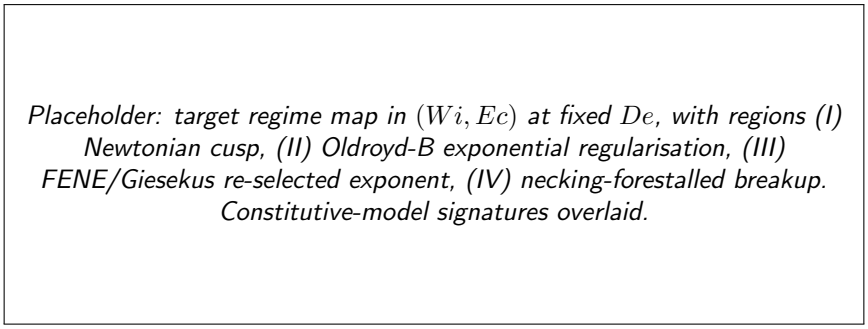


Figure 3: Deliverable. A regime map of viscoelastic singularity classes in (Wi, De, Ec), constructed by numerical sweep of the constitutive ladder. Region (I) is the Newtonian self-similar cusp (Eggers, 1993). Region (II), accessed by Oldroyd-B, is the regularised Entov–Hinch exponential regime in which $R_{\min}$ never reaches zero (Entov and Hinch, 1997). Region (III), set by the finite-extensibility ceiling in Giesekus and FENE-P, is the re-selected singular regime of beads-on-a-string morphology (Bhat et al., 2010). Region (IV), predicted by the Fielding extensional necking criterion (Fielding, 2011; Hoyle and Fielding, 2016) and accessed by Johnson–Segalman and Rolie–Poly, is the regime in which the filament fails through a linear instability of the extensional profile before reaching a self-similar cusp. Shaded overlays indicate where CaBER experiments (McKinley and Tripathi, 2000) already probe published rheology.

What you will do and what you will learn?

The project has four scientific objectives — (i) reproduce the Newtonian self-similar benchmark and the Entov–Hinch exponential regime; (ii) implement a ladder of viscoelastic constitutive models in the log-conformation formulation; (iii) sweep the (Wi, De, Ec) plane and assemble the regime map; (iv) benchmark against CaBER — plus a craft objective (v) of absorbing the CoMPhy-Lab open-source workflow. Items 1–5 below correspond.

1. Reproduce the Newtonian Eggers–Fontelos self-similar pinch-off in Basilisk C as a quantitative benchmark (Eggers, 1993; Eggers and Fontelos, 2015; Popinet, 2015; Sanjay, 2022). Verify the scaling of $R_{\min}$ with $(t_c - t)$ to within a few percent, and recover the Entov–Hinch exponential thinning $R_{\min}(t) \propto \exp(-t/3\lambda)$ for an Oldroyd-B thread (Entov and Hinch, 1997). These two benchmarks calibrate the inertial/viscous Newtonian pinch-off and the elasto-capillary regularisation at opposite ends of the ladder.
2. Implement the constitutive ladder — Oldroyd-B, Giesekus, FENE-P, Johnson–Segalman, Rolie–Poly, and soft-glassy rheology (Bhat et al., 2010; Entov and Hinch, 1997; Likhtman and Graham, 2003; Sollich et al., 1997; Wilson and Fielding, 2006) — as drop-in rheology modules using the log-conformation formulation of the polymer-stress evolution (Fattal and Kupferman, 2004). This keeps the conformation tensor positive-definite through the exponential-thinning regime and into the re-selected singular regime, which is essential for integrating the thread all the way to the last resolvable neck radius. Each module is validated in isolation against published analytical limits before being used in a sweep.
3. Sweep the (Wi, De, Ec) plane at fixed Oh for each constitutive model in the ladder, and locate the boundaries between the three distinct singularity fates: regularisation (Oldroyd-B), re-selected singular exponent (Giesekus / FENE-P beads-on-a-string (Bhat et al., 2010)), and necking-forestalled breakup when the Fielding criterion is satisfied (Fielding, 2011; Hoyle and Fielding, 2016) (Johnson–Segalman, Rolie–Poly, and soft-glassy rheology). Deliver a single regime map showing which model lands in which region, how the boundaries move with finite-extensibility parameters, and where biological and industrial polymer fluids sit.

4. Benchmark the thin-filament simulations against the CaBER methodology (McKinley and Tripathi, 2000): compare the predicted mid-filament radius evolution $R_{\min}(t)$ against published CaBER data for at least one polymer system in each of the four regime-map regions. The CaBER comparison is the project's experimental-contact objective: it tells you, for a given real polymer, which region of the map its rheology actually occupies, and what that implies about the chosen constitutive closure.
5. Learn the full open-source workflow of the CoMPhy Lab: Basilisk setup, adaptive mesh refinement targeted at the thinning neck, Python post-processing, GitHub documentation, and writing for a physics audience (Popinet, 2015; Sanjay, 2022). Deliver a public GitHub repository containing all constitutive-model modules, the sweep driver, and the post-processing pipeline, co-supervised with Prof. Suzanne Fielding's group at Durham.

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